MARY C. BANNER

Fox Island, WA | 253-313-6849 | NDNChik32@gmail.com
LinkedIn: linkedin.com/in/chefmaryb88
Portfolio (GitHub): github.com/BanBan2488

EDUCATION

M.P.S., Environmental Science (Ecosystems: Land, Water and Air)
State University of New York College of Environmental Science and Forestry, Syracuse, NY 2023 – 2025

B.S., Native Environmental Science, Geographic Informational Systems (GIS)
Northwest Indian College, Bellingham, WA 2020 – 2022

A.A, Associate of Arts, Sociology
Pierce College, Fort Steilacoom, WA 2017 - 2019

TECHNICAL SKILLS

GIS & Mapping: ArcGIS Pro, ArcGIS Online, StoryMaps, Survey123, cartographic design, spatial analysis
Remote Sensing: Landsat, Sentinel-2, Earth observation data processing, SeaDAS
Programming & Analysis: Python (ArcPy, Jupyter Notebooks), R (tidyverse)
Data & Spatial Analysis: Raster and vector workflows, NetCDF data processing, GPS data collection
Data Sources & Platforms: NASA Earthdata, ORNL DAAC, USGS EarthExplorer & National Map, NOAA datasets
Other Tools: Adobe Acrobat, AutoCAD, HOMER Pro

PROFESSIONAL EXPERIENCE

Water System Operator in Training
Fox Island Water Mutual Association — WA July 2025 – April 2026

- Operate and maintain public water infrastructure (wells, pumps, distribution systems) supporting reliable community service
- Supported tracking and organization of system data for infrastructure monitoring and reporting
- Conduct water quality sampling and ensure compliance with WA DOH and EPA standards
- Diagnose system failures and implement preventive maintenance strategies
- Coordinate with contractors, board members, and community stakeholders
- Respond to emergency service disruptions and infrastructure failures
- Maintained and organized system records, with interest in developing GIS-based solutions for asset tracking

RESEARCH AND GIS EXPERIENCE

NASA ABoVE Internship – NASA Goddard Space Flight Center 2024–2025

- Produced regional GIS maps for Arctic-Boreal datasets across six Indigenous regions
- Translated complex Earth observation data into plain-language geospatial products
- Contributed to published community-facing NASA booklet
- Bridged scientific datasets and Indigenous knowledge systems

NASA Ames Indigenous Peoples Initiative – NASA Ames Research Center 2023

- Authored white paper using remote sensing data
- Applied Two-Eyed Seeing framework to integrate knowledge systems
- Evaluated web-based GIS tools for environmental monitoring

NASA ABoVE Internship 2022

- Developed ArcGIS StoryMap synthesizing Arctic-Boreal environmental research
- Created interactive GIS products for stakeholder engagement
- Performed spatial analysis and data synthesis

Salish Sea Research Center Internship – Partnership with NASA Ames Research Center 2021

- Conducted GIS analysis of pipeline impacts on wild rice ecosystems
- Identified environmental "hotspots" affecting water quality and food sovereignty
- Produced research poster and presented findings

Yakama Nation Fisheries Internship – Yakama Nation Fisheries 2020

- Conducted fish population surveys and habitat assessments
- Collected and processed ecological and genetic data
- Supported fisheries management and restoration activities

PROFESSIONAL TRAINING AND WORKSHOPS

- **NASA ARSET** – Remote sensing for water quality and watershed analysis, assessing the impacts of fires on watershed health, hyperspectral data for land and coastal systems
- **NASA HYR-SENSE** – Hyperspectral and thermal remote sensing for environmental justice
- **NASA UNBOUND** – Tribal food, energy, and water systems using Earth observation data
- **IndigiData** – Indigenous data science and community-centered applications

AWARDS

- Sloan Indigenous Scholar 2023–2025
- Native Forward Scholar 2025
- S-STEM Scholarship, Northwest Indian College 2021–2022
- Sara Ann Bilezikian Fellowship 2022–2024

PROFESSIONAL ENGAGEMENT AND AFFILIATIONS

Memberships

- American Geophysical Union 2022–Present
- American Indian Science and Engineering Society 2019–Present
- Tribal GIS 2023–Present

Professional Service

- Co-Founder, Indigenous@Pierce Student Club
- Co-Organizer, Native American Heritage Month events
- Co-Organizer, MMIW awareness event

Languages

- English (Native)
- American Sign Language (Intermediate)
- Spanish (Basic)
- Ponca (Basic)
- Anishinaabemowin (Basic)